

The Architecture Principle

A Structural Foundation for Green-Function Theory

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Abstract

Green functions occupy a central position throughout modern mathematical physics. They describe propagation in differential equations, quantum mechanics, quantum field theory, condensed-matter systems, statistical mechanics, inverse problems, and graph theory. During the development of Green-function theory, a rich mathematical hierarchy has emerged, including Hamiltonians, Green functions, Dyson equations, self-energies, Luttinger–Ward functionals, and thermodynamic potentials. Although these mathematical structures have been developed independently for different purposes, they fit together with remarkable coherence. The present work proposes the **Architecture Principle**, according to which these objects are successive mathematical representations of a more primitive entity, termed a **Green-Function Architecture**. Every admissible Green function is interpreted as one representation of an underlying transport architecture characterized by propagation, interaction, holonomy, and recursive closure. These four structural sectors are summarized by a reduced architectural coordinate tuple (a, b, c, d) , which serves as a compact descriptor of architectural complexity. Rather than replacing established Green-function theory, the proposed framework provides a structural interpretation unifying graph theory, spectral graph theory, quantum graphs, inverse problems, topology, and recursive Green-function constructions within a common mathematical language. The work establishes the mathematical foundations of Green-Function Architecture Theory, formulates precise representation principles, proves reduced structural classification results, and identifies the principal open mathematical problems required for a complete reconstruction theory.

Keywords: Green functions; mathematical physics; graph theory; quantum graphs; reconstruction theory; spectral graph theory; topology.

1. Introduction

Green functions have become one of the universal mathematical tools of theoretical physics. Since the pioneering work of George Green in 1828, Green-function methods have been applied to boundary-value problems, differential equations, quantum mechanics, quantum field theory, condensed-matter physics, statistical mechanics, inverse problems, and graph theory [1–9].

During the last century, Green-function theory has evolved far beyond the original concept of a propagator. Modern formulations include Hamiltonian theory, Dyson equations, self-energy formalisms, Luttinger–Ward functionals, Green functions on graphs, spectral graph theory, and recursive diagrammatic expansions [4–18]. Despite this remarkable mathematical richness, these structures are usually introduced independently and for different physical purposes.

The present work is motivated by a simple but fundamental question:

Why does Green-function theory possess such a remarkably coherent mathematical hierarchy?

Rather than considering Green functions as isolated analytical kernels, we propose that they should be regarded as representations of a deeper structural object. This leads to the central hypothesis of the present work.

Architecture Principle.

The primitive mathematical object of Green-function theory is not the Green function itself but a Green-Function Architecture describing the global organization of transport. Green functions, self-energies, functional representations, and statistical descriptions arise as successive mathematical representations of this underlying architecture.

This viewpoint does not modify the established mathematical formalism of Green-function theory. Instead, it proposes a different logical ordering of its principal mathematical objects.

The resulting framework, termed **Green-Function Architecture Theory (GFAT)**, synthesizes ideas originating from graph theory, graph Laplacians, quantum graphs, spectral graph theory, algebraic topology, inverse graph problems, and recursive Green-function methods into a common architectural language.

2. The Mathematical Hierarchy of Green-Function Theory

One of the remarkable achievements of twentieth-century mathematical physics is the development of a hierarchical Green-function formalism.

Beginning with the microscopic Hamiltonian

$$H,$$

one constructs the interacting Green function

G .

The Green function satisfies the Dyson equation, introducing the self-energy

Σ .

The self-energy itself is generated by the Luttinger–Ward functional

$\Phi[G]$,

while thermodynamic quantities follow from the grand potential

Ω .

Consequently, modern Green-function theory naturally possesses the hierarchy

$$H \rightarrow G \rightarrow \Sigma \rightarrow \Phi \rightarrow \Omega.$$

Each level of this hierarchy has been developed rigorously and plays a distinct role within many-body theory [4–12].

From a structural point of view, however, an important observation emerges.

The objects

$$H, G, \Sigma, \Phi, \Omega$$

are mathematically different, yet they are not independent. Each is constructed from the previous level by a well-defined mathematical procedure.

This raises a natural question.

Is there a more primitive mathematical object from which the entire hierarchy itself emerges?

Green-Function Architecture Theory proposes an affirmative answer.

3. The Architecture Principle

The central hypothesis of Green-Function Architecture Theory is that the hierarchy of Green-function theory is itself a representation hierarchy.

Instead of beginning with the Hamiltonian, we introduce a more primitive object,

$$\mathcal{A},$$

called a **Green-Function Architecture**.

The Green-Function Architecture represents the complete structural organization of transport before any analytical representation is chosen.

The Architecture Principle therefore proposes the following ordering:

$$\mathcal{A} \rightarrow H \rightarrow G \rightarrow \Sigma \rightarrow \Phi \rightarrow \Omega.$$

Within this viewpoint, Hamiltonians, Green functions, self-energies, and thermodynamic potentials are interpreted as successive mathematical representations of the same underlying architecture.

The Green function is therefore no longer regarded as the primitive mathematical object. Instead, it becomes one representation of a deeper transport structure.

This interpretation is compatible with several established mathematical frameworks.

Graph Laplacians describe transport through weighted graphs [13–15].

Quantum graphs describe Green functions by sums over propagation paths and vertex scattering processes [16–18].

Inverse graph problems demonstrate that graph structure may often be reconstructed from boundary or spectral information [19–21].

Recursive Green-function constructions organize perturbative expansions through families of Feynman graphs and Hopf-algebraic recursion [22–25].

These developments strongly suggest that Green functions already contain substantial structural information beyond point-to-point propagation.

The Architecture Principle elevates this observation to the level of the primitive mathematical object.

4. Green-Function Architectures

We now introduce the fundamental definition of the theory.

Definition 1 (Green-Function Architecture).

A Green-Function Architecture is the ordered triple

$$\mathcal{A} = (\Gamma, U, G),$$

where

- Γ denotes a directed transport graph,
- U denotes the family of transport operators acting on the graph,
- G denotes the associated Green-function representation.

The graph describes the transport topology.

The transport operators describe propagation through the graph.

The Green function provides the analytical representation of the resulting transport process.

Unlike conventional Green-function theory, where the Green function itself is regarded as the fundamental object, the present framework regards the architecture as primitive and the Green function as one of its mathematical representations.

This interpretation immediately suggests that the structural complexity of a Green-Function Architecture should be describable independently of any particular analytical representation.

The remainder of the present work develops such a reduced structural description based on four primitive architectural sectors.

5. The Four Primitive Architectural Sectors

The central hypothesis of Green-Function Architecture Theory is that every Green-Function Architecture possesses four independent structural components describing distinct aspects of transport organization.

Rather than introducing these sectors axiomatically, we motivate them by observing that modern Green-function theory already separates naturally into four complementary mathematical structures.

The first concerns **propagation** itself. Green functions describe transport between points, while graph Laplacians and quantum graphs describe transport through networks [13–18].

The second concerns **interaction**. In many-body theory interactions are represented by the self-energy, while in graph-based descriptions they correspond to scattering vertices or branching points [4–12,16–18].

The third concerns **global topology**. Green functions depend not only on local propagation but also on the existence of closed transport cycles, graph topology, homology classes, and spectral properties of the underlying transport network [25].

The fourth concerns **recursive closure**. Dyson equations, recursive Green-function methods, Hopf algebras of Feynman graphs, and Dyson–Schwinger equations all demonstrate that Green-function theory possesses an intrinsic recursive organization [22–24].

Accordingly, we define four primitive architectural sectors.

Propagation

The propagation sector contains the transport paths of the architecture together with the corresponding transport operators. It describes how information propagates through the transport graph.

Interaction

The interaction sector contains the structural organization of branching and merging events. It describes how different transport channels communicate through interaction vertices.

Holonomy

The holonomy sector describes global topological organization. Closed transport cycles, independent loops, cycle spaces, and related topological invariants belong to this sector.

Recursive Closure

The recursive sector describes structural feedback. Recursive graph insertions, strongly connected components, recursive transport, and graph self-similarity belong to this sector.

These four sectors are proposed as independent structural components of the architecture.

6. Architectural Coordinates

The structural complexity of the architecture is summarized by the reduced coordinate tuple

$$(a, b, c, d),$$

where

- a denotes propagation complexity,
- b denotes interaction complexity,
- c denotes holonomy complexity,
- d denotes recursive complexity.

The tuple is interpreted as a coordinate system on the space of reduced Green-Function Architectures.

Instead of recording every microscopic feature of the architecture, the tuple records only those structural characteristics that remain invariant under reduced architectural equivalence.

The four coordinates therefore provide a compact structural description analogous to topological invariants, graph invariants, or homological classification schemes.

Within the reduced theory, two Green-Function Architectures are considered equivalent whenever their four architectural coordinates coincide.

This motivates the following definition.

Definition 2 (Reduced Architectural Equivalence).

Two Green-Function Architectures

$$\mathcal{A}_1 \text{ and } \mathcal{A}_2$$

are said to be reduced architecturally equivalent if

$$(a, b, c, d)_1 = (a, b, c, d)_2.$$

The tuple therefore serves as the reduced coordinate system of Green-Function Architecture Theory.

7. Reduced Structural Classification

The coordinate tuple possesses two important mathematical properties.

Theorem 1 (Reduced Completeness).

Within the reduced four-sector model, the tuple

$$(a, b, c, d)$$

completely classifies Green-Function Architectures up to reduced architectural equivalence.

Proof.

Reduced architectural equivalence is defined by equality of the four primitive structural functionals corresponding to propagation, interaction, holonomy, and recursive closure.

Consequently,

$$(a, b, c, d)_1 = (a, b, c, d)_2$$

if and only if the two architectures possess identical values of all four structural functionals.

Therefore, they are reduced architecturally equivalent.

Conversely, reduced architectural equivalence immediately implies equality of the coordinate tuple.

Hence the tuple completely classifies reduced Green-Function Architectures.

A second result concerns minimality.

Theorem 2 (Reduced Minimality).

The coordinate tuple

$$(a, b, c, d)$$

is minimal.

Proof.

Suppose one coordinate is removed.

If the propagation coordinate is omitted, architectures differing only in propagation complexity become indistinguishable.

The same argument applies independently to interaction, holonomy, and recursive closure.

Therefore no coordinate can be removed without destroying completeness.

Hence the four-coordinate system is minimal.

These theorems establish the mathematical consistency of the reduced architectural description. They do **not** prove that every Green-Function Architecture in nature is completely described by four sectors; rather, they show that **within the reduced GFAT model**, the chosen coordinates form a complete and minimal classification.

8. Architectural Multiplicity

The coordinate tuple provides a reduced description of architectural complexity.

To obtain a single scalar measure, we introduce the concept of **Architectural Multiplicity**.

Definition 3 (Architectural Multiplicity).

The Architectural Multiplicity is a scalar invariant

$$\Pi = \Pi(a, b, c, d),$$

associated with every reduced Green-Function Architecture.

Its purpose is to summarize the overall structural richness of the architecture in a single quantity.

At the present stage of the theory, the explicit expression

$$\Pi = 2^a 3^b (3\pi)^c A^{d/8}$$

should be regarded as a **candidate invariant** motivated by structural considerations rather than as a derived theorem.

The central mathematical problem is therefore no longer the definition of Π , but its derivation from graph-theoretic, combinatorial, or transport-theoretic principles.

Existing work on spanning trees, graph Green functions, graph zeta functions, spectral graph invariants, and quantum graphs suggests that such a derivation may ultimately be possible [13–18,25–28], although it remains an open problem.

Consequently, the Architectural Multiplicity should presently be interpreted as the principal conjectural invariant of Green-Function Architecture Theory, providing the bridge between reduced architectural complexity and the analytical representations discussed in the following sections.

9. The Architecture Representation Principle

The principal hypothesis of Green-Function Architecture Theory is that the hierarchy of modern Green-function theory is a hierarchy of mathematical representations rather than a hierarchy of independent mathematical objects.

Accordingly, we formulate the following principle.

Architecture Representation Principle

The Green-Function Architecture is the primitive mathematical object. The principal mathematical structures of Green-function theory arise as successive representations of this architecture.

The proposed representation hierarchy is

$$\mathcal{A} \rightarrow (a, b, c, d) \rightarrow \Pi \rightarrow H \rightarrow G \rightarrow \Sigma \rightarrow \Phi \rightarrow \Omega.$$

Each level represents a different mathematical description of the same underlying architecture.

The first mapping assigns the reduced architectural coordinates to the complete architecture.

The second mapping constructs the corresponding Architectural Multiplicity.

The Hamiltonian then provides an operator representation of the architecture.

The Green function represents propagation within this Hamiltonian description.

The self-energy represents interaction effects, while the Luttinger–Ward functional provides a functional representation of the complete interacting system.

Finally, the thermodynamic potential summarizes the statistical properties of the corresponding ensemble.

The Architecture Principle therefore does not replace the mathematical hierarchy of Green-function theory. Instead, it proposes that the hierarchy itself originates from one deeper structural object.

10. Relation to Existing Mathematical Frameworks

One important feature of Green-Function Architecture Theory is that it is compatible with several well-established branches of mathematics.

Graph theory provides the natural description of transport graphs and connectivity [13–15].

Graph Laplacians provide Green functions through inverse or pseudoinverse operators [13–15].

Quantum graph theory represents Green functions as sums over propagation paths connected by vertex scattering matrices [16–18].

Spectral graph theory relates graph structure to spectral invariants and transport operators [13–15].

Algebraic topology describes global transport organization through cycle spaces, Betti numbers, and homology groups [25–28].

Inverse graph problems investigate the reconstruction of graph structure from Green-function or boundary information [19–21].

Recursive Green-function methods, Hopf algebras, Dyson equations, and Dyson–Schwinger recursion describe recursive organization within perturbative quantum field theory [22–24].

Rather than competing with these mathematical frameworks, Green-Function Architecture Theory seeks to unify them by interpreting each as describing one aspect of the same underlying architecture.

Within this interpretation,

- graph theory describes propagation,
- self-energy theory describes interaction,
- topology describes holonomy,
- recursive Green-function theory describes structural closure.

The four architectural sectors therefore emerge naturally from existing mathematics rather than being introduced independently.

11. Principal Open Mathematical Problems

Although the reduced architectural framework developed in the present work is mathematically consistent, several important questions remain open.

Problem 1

Construct a universal reconstruction map

$$\mathcal{R}: G \rightarrow \mathcal{A}.$$

At present, reconstruction has been established only for restricted graph-based settings.

Problem 2

Prove the universal Four-Sector Decomposition Theorem.

Specifically, determine whether every admissible Green-Function Architecture admits a unique decomposition into propagation, interaction, holonomy, and recursive closure.

Problem 3

Derive the Architectural Multiplicity from graph theory.

The present work proposes the Architectural Multiplicity as a structural invariant.

A rigorous derivation from transport-path enumeration, graph invariants, spanning forests, graph zeta functions, or related combinatorial structures remains to be developed.

Problem 4

Establish the universal Architecture Representation Theorem.

Specifically, determine whether every admissible Green-Function Architecture generates the complete hierarchy

$$H \rightarrow G \rightarrow \Sigma \rightarrow \Phi \rightarrow \Omega.$$

Such a theorem would establish Green-Function Architecture as the common structural origin of modern Green-function theory.

These questions define the future mathematical development of Green-Function Architecture Theory.

12. Discussion

The Architecture Principle proposes a structural interpretation of Green-function theory rather than a replacement of its established mathematical formalism.

Modern Green-function theory already possesses an exceptionally rich hierarchy consisting of Hamiltonians, Green functions, Dyson equations, self-energies, Luttinger–Ward functionals, and thermodynamic descriptions.

The present work proposes that these objects should be interpreted as successive representations of one underlying Green-Function Architecture.

This viewpoint naturally unifies concepts originating from graph theory, quantum graphs, algebraic topology, inverse graph problems, spectral graph theory, and recursive Green-function methods.

The reduced architectural coordinates

$$(a, b, c, d)$$

provide a compact description of the structural organization of the architecture, while the Architectural Multiplicity is proposed as the corresponding scalar invariant.

Unlike conventional classification schemes, which classify physical observables, Green-Function Architecture Theory classifies the transport architecture itself.

The resulting framework establishes a new structural viewpoint from which Green-function theory may be understood.

13. Conclusions

The present work introduces the Architecture Principle as a structural foundation for Green-function theory.

Instead of regarding Green functions as primitive mathematical objects, the proposed framework introduces Green-Function Architectures as the underlying structural entities from which Green functions arise.

The architecture is characterized by four primitive structural sectors corresponding to propagation, interaction, holonomy, and recursive closure.

These sectors are summarized by the architectural coordinate tuple

$$(a, b, c, d),$$

which provides a complete and minimal description within the reduced architectural theory.

The Architectural Multiplicity is introduced as the principal scalar invariant associated with the reduced architecture.

The central contribution of the present work is therefore not a modification of Green-function theory itself but a new interpretation of its mathematical organization.

Green functions, self-energies, functional representations, and thermodynamic quantities are viewed as successive representations of one common architectural object.

This perspective suggests that the remarkable coherence of modern Green-function theory reflects an underlying structural unity rather than a collection of independent mathematical constructions.

Whether the remaining open problems—particularly universal reconstruction, unique four-sector decomposition, derivation of the Architectural Multiplicity, and the Architecture Representation Theorem—can be solved remains an open question. Nevertheless, the Architecture Principle establishes a coherent mathematical framework and defines a precise research program for investigating the structural foundations of Green-function theory.

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Author Contributions

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Ethics Approval

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